

Based on the differences between NiFi 1.x and NiFi 2.x processors and the general steps for RAG (Data Ingestion, Transformation/Chunking, Embedding, and Vector Storage), here is a set of Q&A pairs suitable for feeding your RAG knowledge base.

NiFi 1.x vs. NiFi 2.x RAG Q&A (English)

1. Core Processor Architecture

Question (Q)	Answer (A)	NiFi Context
Q: What is the primary development language for creating custom processors in NiFi 1.x?	A: The primary language for custom processor development in NiFi 1.x is Java (or other JVM-based languages).	NiFi 1.x
Q: What new API and language support did NiFi 2.x introduce for developing native processors?	A: NiFi 2.x introduced a Native Python API , allowing developers to build first-class processors using Python .	NiFi 2.x
Q: In a RAG data pipeline, how would a developer typically integrate a custom Python embedding model in NiFi 1.x?	A: In NiFi 1.x, this was typically done using the ExecuteScript processor (with Jython/Groovy) or ExecuteStreamCommand to call an external Python script, which introduced overhead and complexity.	NiFi 1.x
Q: How does the Native Python API in NiFi 2.x benefit RAG pipelines, specifically for embedding models?	A: The Native Python API allows Python embedding models (using libraries like HuggingFace or OpenAI) to be run directly as a native processor, drastically reducing overhead and simplifying library integration .	NiFi 2.x

Exportar a Hojas de cálculo

2. RAG Pipeline Processors (Ingestion & Transformation)

Question (Q)	Answer (A)	NiFi Context
Q: Which common NiFi processor is used in both versions to fetch documents (like PDFs or JSON) from an S3 bucket for RAG ingestion?	A: The ListS3 and FetchS3Object processors are standard and used in both NiFi 1.x and NiFi 2.x for S3 data ingestion.	Both

Q: In the data preparation stage of RAG, which processor is commonly used to extract text and chunk data from various document types (e.g., PDF, HTML)?	A: Processors like ExtractText or custom processors built around libraries such as Unstructured.io are essential for text extraction and chunking.	Both
Q: Which NiFi 1.x core processor is essential for manipulating the content of a FlowFile, such as modifying text or formatting JSON, before chunking?	A: The ReplaceText processor or the JoltTransformJSON processor are core tools for content manipulation in NiFi 1.x.	NiFi 1.x
Q: NiFi 2.x introduced new features like Git Integration. How does this improve the management of RAG pipelines?	A: The tighter Git-based flow versioning allows data engineers to manage, track, and collaborate on changes to the RAG pipeline logic (e.g., changing chunking strategies) using standard CI/CD practices.	NiFi 2.x

Exportar a Hojas de cálculo

3. Vectorization and Storage

Question (Q)	Answer (A)	NiFi Context
Q: What is the key bottleneck when integrating external vectorization services (like an OpenAI API call) into a NiFi 1.x flow?	A: The primary bottleneck is often the latency and overhead of making external HTTP calls for every chunk using processors like InvokeHTTP .	NiFi 1.x
Q: NiFi 2.x can use native Python processors for embedding. Which type of vector store processor would immediately follow this step?	A: Following the embedding processor, a processor like PutPinecone , PutWeaviate , or a generic PutDatabaseRecord (for pgVector) would be used to store the vectors.	Both/NiFi 2.x
Q: Does the choice of NiFi version (1.x vs 2.x) affect the selection of an external Vector Database like Pinecone or Chroma?	A: No. The choice of external Vector DB is independent of the NiFi version, but NiFi 2.x allows for more efficient data loading via custom Python processors tailored to that specific DB's client library.	Both

Q: What is the main architectural benefit of NiFi 2.x that improves the throughput of high-volume RAG ingestion pipelines?	A: NiFi 2.x features enhanced performance architecture , including better thread management and load balancing , which results in reduced latency and improved horizontal/vertical scalability for processing high-volume data streams.	NiFi 2.x
---	---	-----------------

Exportar a Hojas de cálculo

4. Security and Usability

Question (Q)	Answer (A)	NiFi Context
Q: What framework updates in NiFi 2.x have a positive impact on security and require consideration during migration?	A: NiFi 2.x features major framework upgrades, including Java 21 , Spring 6 , and Jetty 12 , which enhance security but can introduce breaking changes, particularly with mTLS configuration.	NiFi 2.x
Q: What feature in the NiFi 2.x User Interface (UI) improves the developer experience when managing complex RAG flows?	A: The completely rebuilt UI in NiFi 2.x offers faster rendering of complex flows and an improved, more intuitive interface (including features like Dark Mode).	NiFi 2.x